



An Ingestion-based Chronic Internal Radiation Model in *Caenorhabditis elegans* Using Americium Trichloride Reveals Tissue-specific Oxidative Stress and Reproductive Toxicity

Qingyu Xiong ^a, Yi Tan ^a, Chuangcai Li ^a, Xianyu Wu ^a,
Xiaoran Chen ^a and Chun Chen ^{a*}

^a China Jiliang University, College of Life Sciences, Hangzhou, 310018, China.

Authors' contributions

This work was carried out in collaboration among all authors. Authors QX and YT contributed equally to this study. Author CC conceived and designed the study. Authors QX and YT collected the data. Authors XW and CL performed the analysis. Authors CC and XC wrote the manuscript and performed the experiments. All authors read and approved the final manuscript.

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Abstract

Background: Long-term internal exposure to trace environmental radionuclides poses significant health risks. This is especially true for those emitting high linear energy transfer (LET) alpha particles. Traditional radiotoxicology assessments rely heavily on in vitro cultures, mammalian models, or external irradiation systems. A critical research gap exists for chronic internal exposure models of low-penetrating alpha radiation. Such models must use biological uptake pathways to ensure accurate microdosimetry.

*Corresponding author: E-mail: aspring@cjl.u.edu.cn;

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Aims: We aimed to evaluate the biological effects of chronic internal exposure to alpha radiation. To achieve this, we established a chronic ingestion and injury model in *Caenorhabditis elegans* (*C. elegans*). We used americium trichloride (AmCl_3) as the radiation source.

Methodology: We created an ingestion pathway using a liquid exposure system with inactivated *Escherichia coli* (*E. coli*) OP50 and neutralized AmCl_3 (pH 7.0). Age-synchronized L4 nematodes were continuously cultured for 1 to 3 days in a neutral environment. The single-well exposure dose was tightly controlled at 0.748 microsieverts (μSv). We quantified oxidative stress, neurally regulated chemotaxis, and reproductive toxicity. This was achieved using behavioral assays and transgenic fluorescent reporter strains (CF1553, CL2166, PD4251, RW1596).

Results: We discovered highly tissue-specific damage under this extremely low-dose exposure. The expression of antioxidant genes (*sod-3* and *gst-4*) showed a time-dependent, biphasic fluctuation. This process included stress activation, compensatory adaptation, and eventual decompensation. Conversely, body wall muscle structure and chemotaxis function remained normal. This indicates robust radioresistance in these highly differentiated tissues. Our key finding is the pronounced radiosensitivity of the reproductive system. While the basal brood size remained unaffected, the embryo hatching rate decreased significantly ($p < 0.001$). This was accompanied by localized protruding vulva malformations.

Conclusion: This in vivo ingestion-based model effectively reveals the damage induced by endogenous alpha emitters. The observed damage is both time-dependent and tissue-specific. This framework provides a robust biological platform. It can elucidate the toxicological mechanisms of internal radiation and help screen potential radioprotective interventions.

Keywords: *Caenorhabditis elegans*; alpha radiation; oxidative stress; reproductive toxicity.

1. Introduction

The rapid advancement of nuclear energy, nuclear medicine, and deep space exploration has increased the risks of acute and chronic radiation exposure to humans and the biosphere. Consequently, the biological effects of radiation and the underlying protective mechanisms have become focal points in radiobiology and environmental science (Zhu et al., 2026). Compared to acute, high-dose external irradiation, long-term internal exposure to environmental trace radionuclides more accurately reflects real-world radiation contamination scenarios (Zhu et al., 2026). Actinides such as americium-241 (^{241}Am) are of particular concern due to their long half-lives and primary emission of high linear energy transfer (LET) alpha particles during decay. Their preferential accumulation in specific tissues causes severe cellular and tissue damage, making them critical targets for internal microdosimetry and radiotoxicology research (Hofmann et al., 2020; Keenan et al., 2022). Because the weak penetration of ^{241}Am alpha emissions often falls within regulatory exemption limits, its safe use in conventional laboratories is highly feasible, establishing it as an ideal source for internal radiation studies (Zhang, et al., 2022).

Currently, radiotoxicology assessments predominantly rely on in vitro cell cultures or higher mammalian models. The former cannot evaluate systemic tissue-organ interactions, while the latter are limited by long experimental cycles, high costs, and ethical constraints (Zhang et al., 2025). *Caenorhabditis elegans* (*C. elegans*), characterized by a short life cycle, well-characterized genetic background, and fixed somatic cell lineage, has emerged as a prominent model for toxicological and biological assessments (Xu et al., 2026). Multiple studies demonstrate that *C. elegans* is not only a highly reliable bioindicator for assessing the toxicity of environmental chemical mixtures, heavy metals, and nanomaterials (Wittkowski et al., 2019), but also an ideal in vivo model for studying ionizing radiation damage (Yuan et al., 2023). For instance, extensive research has investigated DNA damage, multigenerational reproductive toxicity, and proteomic alterations in *C. elegans* induced by chronic gamma or X-ray irradiation (Guédon et al., 2021; Dhakal et al., 2021). Nevertheless, existing nematode-based radiation models are primarily limited to highly penetrating external irradiation systems. A significant research gap remains regarding alpha particles, which possess weak penetration but exceptionally high ionization density, particularly concerning chronic internal radiation models based on biological uptake pathways (Zhu et al., 2026).

To address this gap and establish a systematic evaluation method for multidimensional, tissue-specific internal exposure damage, this study develops and validates a chronic internal radiation model in *C. elegans* using americium trichloride as the radiation source. By facilitating the biological uptake of ^{241}Am compounds, we

construct an Exemption-level microdosimetry and environmentally stable experimental framework to systematically evaluate the Oxidative stress disruption, neural chemotaxis maintenance, and embryonic developmental arrest of chronic internal alpha-particle exposure.

2. Materials and Methods

2.1 Main Reagents and Experimental Materials

2.1.1 Americium Trichloride (AmCl_3) and Radiation Dose Measurement

Americium trichloride (AmCl_3 aqueous solution), used to establish the chronic internal radiation model, was provided by the National Institute of Metrology, China (Catalog number: RYAM241, Specification: 10 g/bottle, Calibrated activity: 1.88×10^3 Bq). The radiation emitted by this source falls within regulatory exemption limits, permitting its safe and compliant use in standard laboratory settings. To assess safety and exposure levels, the radiation dose was calculated using the following formula: $D_R = M_L \times A_S \times F_C$, where D_R is the radiation dose, mSv; M_L is the liquid mass, g; A_S is the specific activity, Bq/g; F_C is the dose conversion factor, mSv/Bq.

Based on this calculation, the total radiation dose generated by 10 g of AmCl_3 is 0.374 mSv. This value is significantly below the skin irradiation limit (50 mSv) stipulated in General Administration of Quality Supervision, Inspection and Quarantine of the People's Republic of China (2002), thereby ensuring the exemption-level microdosimetry of all experimental operations.

2.2 Nematode Strains and Synchronization

This study utilized the wild-type N2 strain and four transgenic fluorescent reporter strains: CF1553, CL2166, PD4251, and RW1596. Their primary biological characteristics are detailed in Table 1. Nematodes were routinely maintained in a 20 °C incubator on Nematode Growth Medium (NGM) agar plates seeded with the uracil-auxotrophic *Escherichia coli* (*E. coli*) OP50 strain (Brenner, 1974). Age-synchronized L4 larvae were utilized for all subsequent exposure experiments.

Table 1. Introduction to nematode strains

Strains	Description
CF1553	[(pAD76) <i>sod-3p::GFP</i> + <i>rol-6(su1006)</i>]. This strain is associated with superoxide dismutase activity and is one of the oxidative stress-related strains.
CL2166	[(pAF15) <i>gst-4p::GFP::NLS</i>] This strain is associated with glutathione transferase activity and is one of the oxidative stress-related strains.
PD4251	[(pSAK2) <i>myo-3p::GFP::LacZ::NLS</i> + (pSAK4) <i>myo-3p::mitochondrial GFP</i> + <i>dpy-20(+)</i>] I. This strain produces GFP in all body wall muscles and vulval muscles, with a combination of mitochondrial and nuclear localization.
RW1596	[<i>myo-3p::GFP::myo-3</i> + <i>rol-6(su1006)</i>]. This strain has GFP localized to myosin within the body wall muscles, which therefore allows visualization of the sarcomeres.
N2	Wildtype

2.3 Buffer and Media Preparation

M9 buffer: The buffer contained 6 g/L Na_2HPO_4 , 3 g/L KH_2PO_4 , 5 g/L NaCl, and 0.25 g/L $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. The volume was adjusted to 1 L with deionized water, and the solution was sterilized by autoclaving at 121 °C for 20 min prior to use.

S medium: Basal solution: 5.85 g NaCl, 1 g K_2HPO_4 , 6 g KH_2PO_4 , and 1 mL cholesterol (5 mg/mL in absolute ethanol) were dissolved, adjusted to 1 L with deionized water, and autoclaved.

1 M Potassium citrate buffer (pH 6.0): 20 g citric acid and 293.5 g tripotassium citrate· H_2O were dissolved in water, adjusted to 1 L, and autoclaved.

Trace metals solution: 1.86 g disodium EDTA, 0.69 g $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, 0.2 g $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$, 0.29 g $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$, and 0.025 g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ were dissolved in water, adjusted to 1 L, autoclaved, and stored in the dark.

1 M CaCl₂ solution: 55.5 g CaCl₂ was dissolved in water, adjusted to 1 L, and autoclaved.

Working medium preparation: To prepare the working S medium, 10 mL of 1 M potassium citrate buffer (pH 6.0), 10 mL of trace metals solution, 3 mL of 1 M CaCl₂, and 3 mL of 1 M MgSO₄ were aseptically added to 1 L of the autoclaved basal solution and mixed thoroughly. No further autoclaving was required.

2.4 OP50 Culture and Inactivation Treatment

Live OP50 bacteria can accelerate their proliferation upon absorbing lanthanide-series elements, which interferes with nematode growth and the toxicity assessment of americium compounds (Singer et al., 2023). To prevent this, OP50 was inactivated using a two-step procedure. First, collected OP50 bacterial pellets were pre-frozen at -80 °C and subsequently lyophilized for 48 h to produce a dry powder. This OP50 lyophilized powder was then subjected to high-dose ionizing radiation sterilization at a specialized facility, ensuring complete bacterial inactivation while preserving cellular morphology.

2.5 Construction and Optimization of the Americium Trichloride Exposure System

2.5.1 Dilution and Neutralization of the Americium Solution

To provide a 72-h continuous ingestion exposure via inactivated OP50 in a neutral microenvironment (pH 7.0) for the chronic internal radiation exposure of *C. elegans*, the highly acidic americium trichloride (AmCl₃) stock solution was neutralized and standardized. For the 96-well plate exposure system, the final volume per well was established at 200 µL (comprising 90 µL of S medium, 10 µL of OP50 bacterial suspension, and 100 µL of the neutralized AmCl₃ solution).

2.5.1.1 Neutralization and Volumetric Dilution of Americium Trichloride

One milliliter of the AmCl₃ stock solution was accurately transferred to a glass vial and weighed (recorded as x, in g). The solution was slowly titrated with 1 mol/L NaOH to adjust the pH to 7.0 (with the volume of NaOH consumed recorded as z, in mL). If over-titration occurred, an HCl solution was used to restore neutrality (recorded as z1, in mL). Following neutralization, the total volume was adjusted to 2.5 mL with deionized water to ensure uniform and stable radiation doses across different experimental batches.

2.5.1.2 Radioactivity and Dose Calculation

The final Effective radiation dose (M100, µSv) per well of working solution (containing 100 µL of americium solution) is directly calculated using the following integrated formula.

$$M_{100} = \frac{0.1 \times 187 \times x \times A}{2.5} = 7.48x \cdot A$$

(Where x is the mass of 1 mL of the stock AmCl₃ solution, g; 187 is a constant based on specific activity; A is the dose conversion factor for AmCl₃; 2.5 is the final constant volume after neutralization, mL).

2.5.1.3 Final Exposure Dose Calibration

The mass of 1 mL of the AmCl₃ stock solution was determined to be approximately 1 g. Substituting this value into the formulas above, the calculated radiation dose generated by 100 µL of the neutralized and diluted AmCl₃ solution is 1.496 µSv. Upon integration into the 96-well plate system (final volume of 200 µL), the precise radiation exposure dose per well was controlled at 0.748 µSv.

2.5.2 Nematode Liquid Exposure Culture System

Food preparation: The lyophilized OP50 powder was resuspended in S medium, and its optical density at 600 nm (OD₆₀₀) was adjusted to 0.6.

System assembly (Stiernagle, 2006): The exposure system was assembled in 96-well plates by sequentially adding 90 μL of S medium, 10 μL of the OP50 suspension ($\text{OD}_{600} = 0.6$), and 100 μL of the neutralized AmCl_3 working solution, yielding a final volume of 200 μL per well.

Inoculation and culture: Healthy L4 larvae were inoculated into the exposure system (30 worms/well, performed in biological triplicates). To prevent evaporation, peripheral wells surrounding the experimental wells were filled with S medium. The plates were sealed with breathable film and maintained in a 20 °C incubator. To ensure adequate nutrition and oxygenation, 10 μL of fresh OP50 suspension ($\text{OD}_{600} = 0.6$) was supplemented every 48 h, accompanied by brief ventilation.

2.6 Measurement of Toxicological Endpoints

2.6.1 Quantification of Oxidative Stress Levels

Age-synchronized L4 transgenic nematodes (CF1553, CL2166, PD4251, and RW1596) were exposed to AmCl_3 in the 96-well system, alongside control groups treated with equal volumes of S medium. Following 1, 2, and 3 days of exposure, worms were harvested and washed three times with M9 buffer (centrifuged at 3,000 rpm for 1 min). The nematodes were then mounted on a 2% agarose pad and anesthetized using 10 mM levamisole hydrochloride. Epifluorescence images were captured using an Olympus IX73 inverted fluorescence microscope equipped with a 10 $\times$ objective lens (excitation: 488 nm, emission: 510 nm). Exposure times were kept constant across all comparative groups to ensure quantitative reliability. Fluorescence quantification was performed using ImageJ software (National Institutes of Health). For each nematode, the whole-body Mean Gray Value was measured, and the background fluorescence was subtracted to determine the relative fluorescence intensity.

2.6.2 Chemotaxis Behavior Analysis

Following AmCl_3 exposure (or control treatment), nematodes were collected, washed twice with S medium, and transferred to OP50-seeded NGM plates for a 2-h recovery period. Chemotaxis assay plates (sterile NGM) were prepared as described by Bargmann *et al.* (1993), and illustrated in Fig. 1. A central inoculation zone with a 0.5 cm radius was demarcated, and the plate was divided into four quadrants (A, B, C, and D). An attractant (100 mM NaCl) was spotted at diametrically opposed quadrants (A and C), while a control solution (2 μL of M9 buffer and an equivalent volume of levamisole) was spotted in the remaining quadrants (B and D).

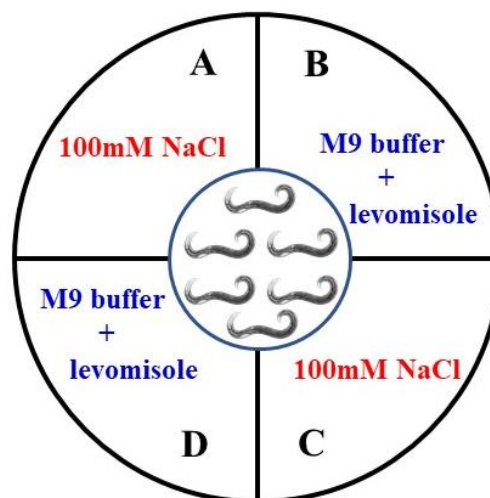


Fig. 1. Diagram of chemotaxis assay plate preparation

Post-recovery, the nematodes were washed two to three times with S medium to remove residual bacteria, and approximately 50–100 worms were placed into the central inoculation zone. After a 1-h incubation at 20 °C, the number of nematodes in each quadrant was recorded. Each independent experiment included three technical

replicates and was performed in three independent biological replicates. The Chemotactic Index (CI) was calculated as follows:

$$CI = \frac{(A+C)-(B+D)}{T}$$

(Where A, B, C, D are the numbers of worms, T is the total worms on the plate). CI approaching +1 indicates strong attraction, whereas a CI approaching -1 indicates strong repulsion.

2.6.3 Embryonic Lethality Rate Assessment

Age-synchronized L1 larvae, obtained via alkaline lysis, were cultured for 12 h and subsequently transferred to the AmCl₃ exposure system for 24 h. Post-exposure, the larvae were washed twice with S medium and transferred to standard NGM plates for recovery. Ten larvae per group that successfully developed to the L4 stage were individually transferred to OP50-seeded NGM plates for oviposition. After 24 h, the parent worms were removed, and the total number of eggs laid per plate was recorded. The plates were then incubated at 20 °C for an additional 24 h, after which successfully hatched larvae were counted using a stereomicroscope. This experiment was performed in three independent biological replicates.

2.7 Statistical Analysis

All statistical analyses were conducted utilizing DPS software. Data visualization were conducted using GraphPad Prism (version 8.0.1). Prior to parametric testing, data were evaluated for normality using the Shapiro-Wilk test and for homogeneity of variance using Levene's test. Continuous variables are presented as the mean ± standard deviation (SD). To compare differences among multiple groups across different time points, a two-way analysis of variance (ANOVA) was performed, followed by Tukey's multiple comparisons test. Statistical significance was defined as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$. All experiments were performed in at least three independent biological replicates.

3. Results

3.1 Oxidative Stress and the Internal Radiation Damage Model

To visualize the spatiotemporal dynamics of oxidative stress, we utilized the CF1553 and CL2166 transgenic reporter strains (Table 1). Compared to the control group, the relative fluorescence intensity of the CF1553 strain was significantly upregulated by 2.34-fold on day 1 (2.34 ± 0.01 , $p = 0.04$; Fig. 2A), 1.92-fold on day 2 (1.92 ± 0.03 , $p = 0.02$; Fig. 2B), and 2.47-fold on day 3 (2.47 ± 0.01 , $p = 0.0004$; Fig. 2C).

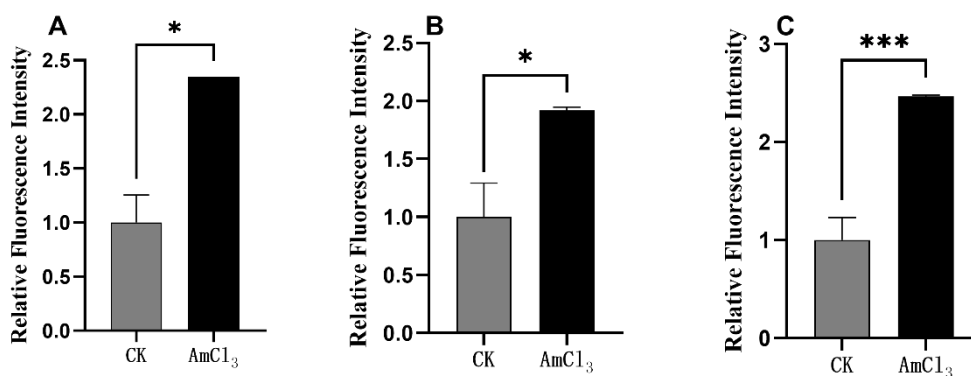


Fig. 2. Relative fluorescence intensity of the CF1553 strain on day 1 (A), day 2 (B), and day 3 (C)

For the CL2166 strain, the relative fluorescence intensity exhibited a gradual upward trend with prolonged exposure. While no significant differences were observed on day 1 (1.08 ± 0.01) and day 2 (1.1 ± 0.1), 1.08- and

1.10-fold relative to the control, respectively; (Fig. 3A, B), the fluorescence intensity significantly increased to 1.13-fold by day 3 (1.13 ± 0.06 , $p < 0.05$; Fig. 3C).

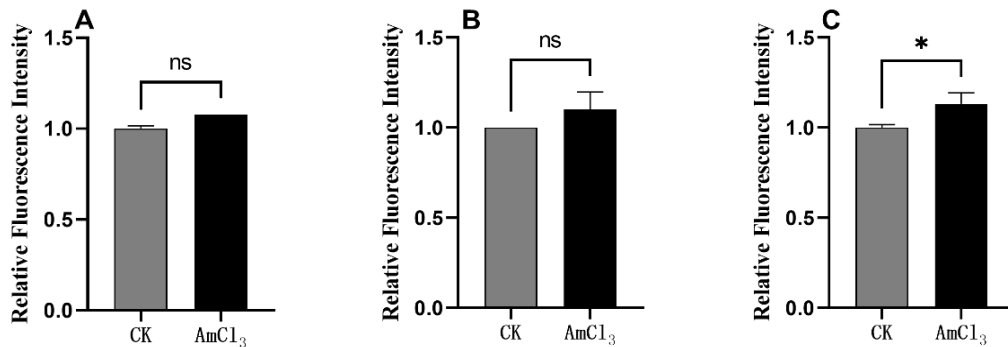


Fig. 3. Relative fluorescence intensity of the CL2166 strain on day 1 (A), day 2 (B), and day 3 (C)

These fluorescent reporter models reveal a complex, biphasic fluctuation in oxidative stress induced by chronic internal americium trichloride (AmCl₃) exposure (Kuzmic et al., 2019). Specifically, the CF1553 strain (reporting *sod-3* expression) exhibited a significant fluorescence increase on day 1, indicating a rapid cellular response to a surge in reactive oxygen species (ROS) and the swift activation of the SKN-1/Nrf2 pathway. The subsequent decline in the fluorescence signal on day 2 suggests a temporary restoration of redox homeostasis by the endogenous antioxidant defense system, marking an adaptation phase. However, by day 3, the highly significant upregulation of fluorescence in both CF1553 and CL2166 (reporting *gst-4* expression) indicates that the accumulated radiation dose likely exceeded the organism's antioxidant compensatory capacity, resulting in secondary ROS accumulation. This transcriptionally evident "stress activation–compensatory adaptation–decompensation" pattern highlights the dynamic response of antioxidant genes to chronic, low-dose radiation.

3.2 Effects of Internal Radiation on the Muscular and Nervous Systems

To assess muscle development and integrity, we utilized the PD4251 and RW1596 strains (Table 1), which express GFP driven by the body wall muscle-specific *myo-3* promoter. For the PD4251 strain, the relative fluorescence intensity of days 1~3 were 1.13 ± 0.32 , 0.73 ± 0.07 and 1.45 ± 0.69 , respectively. For the RW1596 strain, the relative fluorescence intensity of days 1~3 were 1.01 ± 0.06 , 1.27 ± 0.14 and 1.02 ± 0.01 , respectively. Over three consecutive days of observation, no significant differences in relative fluorescence intensity were detected in either strain compared to the control ($p > 0.05$; Fig. 4A–C, Fig. 5A–C). This lack of alteration indicates that AmCl₃ exposure did not induce structural damage (e.g., muscle fiber rupture), abnormal expression of muscle-specific proteins (e.g., myosin), or muscle cell death.

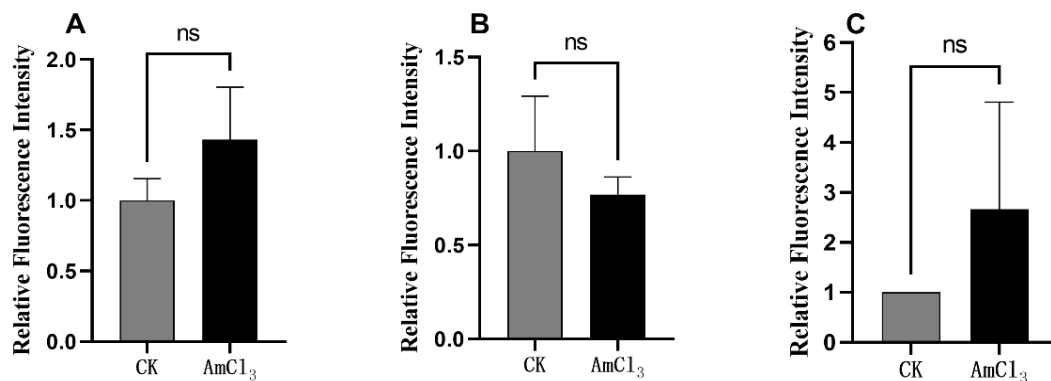


Fig. 4. Relative fluorescence intensity of the PD4251 strain on day 1 (A), day 2 (B), and day 3 (C)

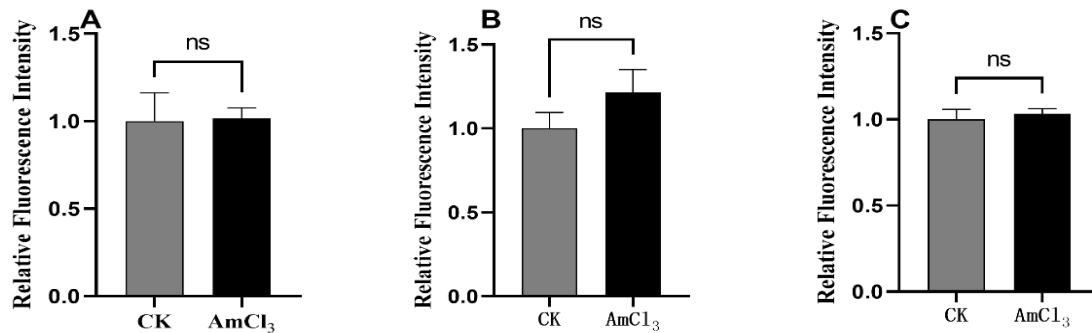


Fig. 5. Relative fluorescence intensity of the RW1596 strain on day 1 (A), day 2 (B), and day 3 (C)

Furthermore, behavioral chemotaxis assays demonstrated that the treated nematodes maintained an intact ability to sense and track NaCl gradients (Fig. 6). This suggests that AmCl_3 exposure did not significantly impair the physiological functions of the nematode nervous system.

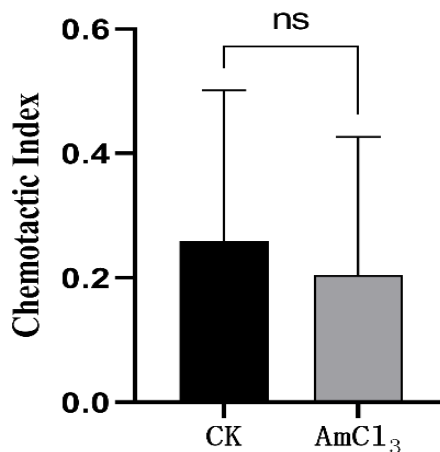


Fig. 6. Chemotaxis Index of nematodes

The absence of observable structural and functional damage in both muscular and nervous tissues confirm the high tissue specificity of chronic, low-dose internal radiation. We hypothesize that highly differentiated nervous and muscle tissues possess strong radioresistance, allowing them to withstand the low-dose ($0.748 \mu\text{Sv}$) chronic exposure employed in this study without manifesting overt structural or functional impairment.

3.3 Effects of Internal Radiation on Reproductive Capacity and Vulval Malformation

AmCl_3 exposure did not significantly alter the basal brood size of the nematodes, as no statistical difference was observed between the treated and control groups ($p > 0.05$; Fig. 7A). However, the exposure caused a sharp and highly significant decrease in the embryo hatching rate ($p < 0.001$; Fig. 7B). Additionally, treated nematodes exhibited protruding vulva malformations (Fig. 8), providing visual, structural evidence of impaired reproductive organ development.

4. Discussion

This study successfully established an experimental framework to evaluate the multidimensional damage effects of chronic internal radiation exposure using an americium compound. Because ^{241}Am primarily emits short-range, high linear energy transfer (LET) alpha particles, conventional external irradiation cannot effectively penetrate the tough cuticle of *C. elegans*. To overcome this, our model utilized a liquid exposure system that

leverages the nematode's continuous pharyngeal pumping behavior. This allows the nematodes to ingest a suspension containing the americium compound and inactivated *E. coli* OP50, thereby establishing an ingestion-based exposure route. Because alpha particles have extremely weak penetration and cannot effectively cross the nematode cuticle from the external environment, the observed tissue-specific damage is strongly indicative of internal exposure following the ingestion of AmCl_3 . However, without direct in vivo micro-distribution imaging, we cannot entirely exclude the potential synergistic contribution of the americium's localized chemical heavy-metal toxicity within the intestinal lumen. While the damage phenotypes align with classic radiotoxicity, distinguishing these dual effects warrants further investigation. Through precise physical dosimetry and pH adjustments, the final environmental exposure dose was standardized at $0.748 \mu\text{Sv}$. This low dose not only ensures laboratory safety by remaining well below national radiation exposure limits, but it also successfully maintains a stable, chronic internal irradiation microenvironment within the nematodes. Furthermore, to prevent experimental confounding caused by the proliferation of host bacteria—which can be stimulated by actinides (Singer et al., 2023)—we inactivated the OP50 using vacuum freeze-drying combined with high-dose irradiation. This critical step ensured the source specificity of the damage phenotypes and enhanced the overall reliability of the model.

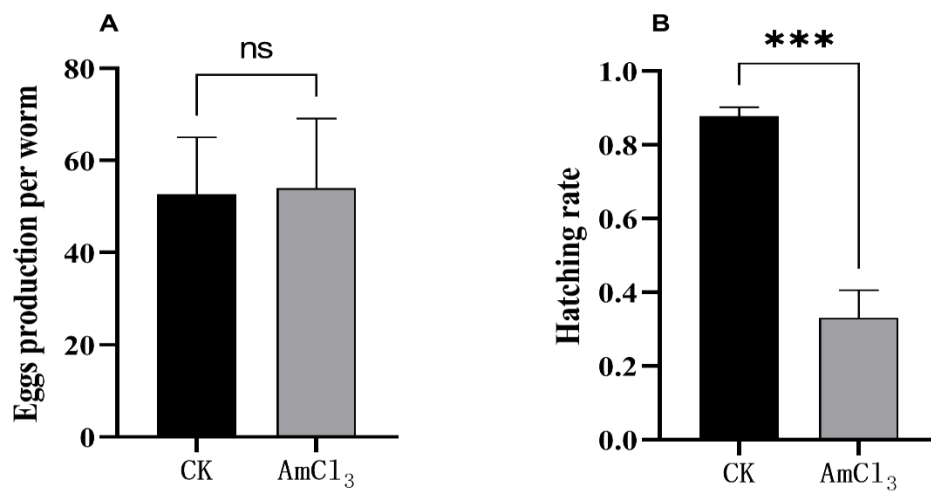


Fig. 7. Bar chart of average brood size (A) and hatching rate (B) under AmCl_3 treatment



Fig. 8. Protruding vulva malformations

Our findings demonstrate that chronic internal radiation induces multidimensional, highly tissue-specific damage in nematodes. As outlined in Table 1, the dual use of the CF1553 (*sod-3*) and CL2166 (*gst-4*) strains

allowed us to distinguish between superoxide dismutase activity and glutathione transferase activity, thereby providing a more comprehensive picture of the SKN-1/Nrf2 pathway activation observed in our results. Regarding oxidative stress, radiation triggered a rapid cellular response on the first day of exposure (evidenced by enhanced *sod-3*-related fluorescence) and exhibited a cumulative effect by the third day, specifically activating the SKN-1/Nrf2-mediated antioxidant homeostasis pathway. Conversely, fluorescence expression in body wall muscle-specific reporter strains (PD4251 and RW1596) remained normal, ruling out non-specific structural muscle damage. Concurrently, the chemotaxis index (CI)—a metric for neurally regulated behavior—showed no significant decline, confirming the high radioresistance of the nematode nervous system.

This disparity in tissue sensitivity is likely directly related to the microenvironment of spatial energy deposition characteristic of internal alpha radiation. Because alpha particles have an extremely short range, their ionization energy is predominantly deposited in the intestinal epithelium (which lacks cuticular protection) and the physically adjacent gonadal tissues (Lay et al., 2019). At a fundamental level, this tissue specificity aligns with the classic Law of Bergonié and Tribondeau (Zhou et al., 2025), which postulates that the radiosensitivity of a tissue is directly proportional to its cellular proliferation rate and inversely proportional to its degree of differentiation (Furukawa et al., 2020). The neurons and body wall muscle cells of nematodes are highly differentiated, post-mitotic terminal cells, thereby exhibiting strong radioresistance to radiation-induced DNA breaks. In contrast, the reproductive system is the most actively proliferating region in the nematode, and its germline stem cell developmental signaling pathways (such as EGF-Ras-MAPK) are highly vulnerable to genotoxic stress (Aklilu et al., 2022).

Although the 0.748 μSv exposure did not significantly reduce the average total brood size of the parent nematodes, it caused a precipitous drop in the embryo hatching rate and induced protruding vulva malformations in some adults. From a developmental perspective, this differential impact on reproductive performance suggests that oogenesis and ovulation were not severely inhibited, allowing for normal brood sizes. Therefore, the extremely low hatching rate clearly indicates a severe disruption during embryogenesis. Low-dose ionizing radiation likely interfered with the early embryonic cellular network during specific mitotic phases (Torfeh et al., 2019). More critically, this profound reproductive failure reflects deep-seated genomic instability. Recent research indicates that parental radiation exposure can cause significant DNA fragmentation in gametes; when these gametes—carrying unrepaired DNA double-strand breaks—unite, they trigger severe chromosomal abnormalities that ultimately lead to embryonic lethality (Wang et al., 2023). Furthermore, the morphological deformities observed in the vulva—a critical structure for egg-laying and development in *C. elegans*—confirm that local tissues suffered profound physical and developmental radiation damage. While the unchanged basal egg production suggests no complete mechanical obstruction to oviposition, the presence of protruding vulval malformations may still exert abnormal physical stress or mechanical damage on the eggs during expulsion, potentially contributing to the unviability of the embryos. Nevertheless, given the known highly genotoxic nature of alpha particles, radiation-induced DNA fragmentation within the gametes and subsequent embryonic genomic instability likely plays a critical, concurrent role in this profound reproductive failure. Future cytological studies are needed to uncouple these structural and genetic factors.

Traditional radiation dosimetry often relies on the whole-body absorbed dose to assess biological risk. However, in this study, an extremely low targeted dose (0.748 μSv) was sufficient to induce severe reproductive failure. This suggests that for internal exposure to radionuclides emitting high-LET radiation, traditional assessment methods based on macroscopic dose averages may fail to accurately capture the true localized biological risk. Therefore, integrating microdosimetry evaluation strategies is essential for accurately assessing the micro-toxicological effects on target organs (Yu et al., 2021).

While this study successfully established a chronic internal radiation model in *C. elegans* using AmCl_3 , several limitations must be acknowledged. First, without direct in vivo micro-distribution evidence from isotope tracing or mass spectrometry imaging, the exact deposition of ^{241}Am in the nematode's intestine or gonads remains unquantified. Second, the model relies entirely on *C. elegans*. While advantageous for genetic studies, the nematode's physiological simplicity limits the direct translation of these findings to higher organisms, as it cannot replicate the complex tissue interactions and systemic immune responses present in mammals (Gabaldón et al., 2024). Third, the current study utilized a single internal radiation dose ($\sim 0.748 \mu\text{Sv}$). Although sufficient to induce detectable changes in ROS levels and tissue morphology, the absence of a dose-response curve limits our understanding of threshold effects and potential adaptive responses at varying exposure levels. Finally, while fluorescent reporter genes (e.g., *sod-3::GFP*) provided insights into oxidative stress activation, future

studies will require comprehensive multi-omics data to elucidate the full spectrum of transcriptional and epigenetic alterations induced by chronic internal radiation.

5. Conclusion

C. elegans serves as a highly effective in vivo model for evaluating the toxicological effects of chronic internal exposure to actinides, such as ^{241}Am . The americium trichloride (AmCl_3) feeding exposure system established in this study offers distinct advantages in safety, experimental controllability, and phenotypic readout, thereby facilitating the precise analysis of tissue-specific and time-dependent damage induced by internal alpha irradiation. Furthermore, this model provides a robust platform for future investigations into tissue-specific radionuclide deposition, the underlying molecular pathways of stress and reproductive toxicity, and the screening of novel radioprotective and chelation therapies.

Disclaimer (Artificial Intelligence)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc) and text-to-image generators have been used during writing or editing of this manuscript.

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Competing Interests

Authors have declared that no competing interests exist.

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